Chapter 7

Normal Subgroups & Quotients

Imagine that you arrive late to algebra class one day, somewhat more tired than usual, and you sit down in back where you hope not to be noticed. You see the professor up front holding a square with a black dot in the center of one side and talking about a new group $\mathbf{D}_4$. But, in your present state and location, you don't see the little dot in the corner. To you the elements of the subgroup $\mathbf{N} := \{e, a, a^2, a^3\}$ all appear to be equivalent and indistinguishable, as do also the elements of the right coset $\mathbf{N}d = \{d, ad, a^2d, a^3d\}$. To you it appears that the professor is presenting a 2-element group $\mathbf{G} = \{N, Nd\}$. When asked to fill out a table for the group, you quickly whip out the a nice little group table below. It has $\mathbf{N}$ as its identity and is isomorphic to $\mathbf{Z}_2$.

$*$	N	Nd
N	N	Nd
Nd	Nd	N

It is only after a few idle minutes while your fellow students continue hard at work that you look over at what they are doing and discover your error. But wait a minute. Haven't you just found a neat way to make a new group from the right cosets of a subgroup?!

Let $\mathbf{G}$ be any group with a subgroup $\mathbf{N}$, and let $\mathbf{G}/\mathbf{N}$ denote the set of right cosets of $\mathbf{N}$ in $\mathbf{G}$. [We use this notation because the *number* of right cosets of $\mathbf{N}$ in $\mathbf{G}$ is $\circ(\mathbf{G})/\circ(\mathbf{N})$ when $\mathbf{G}$ is finite.] Define a binary operation $\cdot$ on $\mathbf{G}/\mathbf{N}$ by

$$\mathbf{N}x \cdot \mathbf{N}y := \mathbf{N}xy \quad \text{for all} \quad x, y \in \mathbf{G}. \tag{$\star\star$}$$

In words, the product of the right coset of x and the right coset of y is the right coset of xy. Notice that $\mathbf{G}/\mathbf{N}$ inherits group properties from $\mathbf{G}$:

- $\cdot$ is associative since $\mathbf{N}x \cdot (\mathbf{N}y \cdot \mathbf{N}z) = \mathbf{N}x \cdot \mathbf{N}yz = \mathbf{N}x(yz) = \mathbf{N}(xy)z = \mathbf{N}xy \cdot \mathbf{N}z = (\mathbf{N}x \cdot \mathbf{N}y) \cdot \mathbf{N}z$,
- $\mathbf{N}e \cdot \mathbf{N}x = \mathbf{N}ex = \mathbf{N}x = \mathbf{N}xe = \mathbf{N}x \cdot \mathbf{N}e$, so $\mathbf{N} = \mathbf{N}e$ is an identity,
- $\mathbf{N}x \cdot \mathbf{N}x^{-1} = \mathbf{N}xx^{-1} = \mathbf{N}e = \mathbf{N}x^{-1}x = \mathbf{N}x^{-1} \cdot \mathbf{N}x$, so $\mathbf{N}x^{-1}$ is the inverse of $\mathbf{N}x$.

By gum, $\mathbf{G}/\mathbf{N}$ is a group! We call $\mathbf{G}/\mathbf{N}$ the **quotient group** of **G modulo N** (or **G mod N** for short).

Problem 82. *List the distinct right cosets of the subgroup* $\mathbf{N} := \{7n \mid n \in \mathbf{Z}\}$ *of* $\mathbf{Z}$ *and construct a table for the quotient group* $\mathbf{Z}/\mathbf{N}$ *(the* **integers modulo** 7*).*

Problem 83. *List the distinct right cosets of the subgroup* $\mathbf{N} := \{0,3,6,9\}$ *of* $\mathbf{Z}_{12}$ *and construct a table for the quotient group* $\mathbf{Z}_{12}/\mathbf{N}$.

Problem 84. *List the distinct right cosets of the subgroup* $\mathbf{N} := \{e,c\}$ *of* $\mathbf{S}_3$ *and construct a table for the quotient group* $\mathbf{S}_3/\mathbf{N}$.

Now, if you were careful you should have run into some difficulty with the last quotient group $\mathbf{S}_3/\mathbf{N}$. What, for example, is the product of the right cosets $\mathbf{N}a = \{a, a^2c\}$ and $\mathbf{N}a^2 = \{a^2, ac\}$? On the one hand, we have

$$\mathbf{N}a \cdot \mathbf{N}a^2 = \mathbf{N}a^3 = \mathbf{N}e = \mathbf{N}.$$

On the other hand, $\mathbf{N}a = \mathbf{N}a^2c$ and $\mathbf{N}a^2 = \mathbf{N}ac$, so that

$$\mathbf{N}a \cdot \mathbf{N}a^2 = \mathbf{N}a^2c \cdot \mathbf{N}ac = \mathbf{N}(a^2c)(ac) = \mathbf{N}a \neq \mathbf{N}!$$

It appears that the "product" of the cosets $\mathbf{N}a$ and $\mathbf{N}a^2$ changes depending on which names (representatives) we choose for these cosets. In a case like this we say here that the "operation" $\cdot$ on $\mathbf{S}_3/\mathbf{N}$ is **not well defined**, that is, it is simply nonsense.

Definition *A subgroup* $\mathbf{N}$ *of a group* $\mathbf{G}$ *is called a* **normal subgroup** *of* $\mathbf{G}$ *if the representative operation* $\cdot$ *on* $\mathbf{G}/\mathbf{N}$ *given by* $(\star\star)$ *is* ***well defined****, that is,*

$$\textit{if } \mathbf{N}x_1 = \mathbf{N}x_2 \textit{ and } \mathbf{N}y_1 = \mathbf{N}y_2, \textit{ then } \mathbf{N}x_1y_1 = \mathbf{N}x_2y_2$$

for all $x_1, x_2, y_1, y_2 \in \mathbf{G}$.

Problem 85. *Verify that* $\mathbf{N} := \{e, a, a^2, a^3\}$ *is a normal subgroup of* $\mathbf{D}_4$.

We can now stop to summarize what we have found.

Theorem *Let* $\mathbf{N}$ *be a subgroup of the group* $\mathbf{G}$. *If* $\mathbf{N}$ *is a normal subgroup, then* $\mathbf{G}/\mathbf{N}$ *is a group with operation* $\cdot$ *defined by* $(\star\star)$. *If* $\mathbf{N}$ *is not normal, then* $(\star\star)$ *does not define a binary operation at all.*

What we need to know, then, is how to tell whether or not a subgroup $\mathbf{N}$ is normal. There are several different ways to do this.

Theorem 86. *For a group* $\mathbf{G}$ *and a subgroup* $\mathbf{N}$*, these are equivalent.*

(i) $\mathbf{N}$ *is a normal subgroup of* $\mathbf{G}$.

(ii) $\mathbf{N}x = x\mathbf{N}$ *for all* $x \in \mathbf{G}$ (*every right coset is a left coset*).

(iii) *For all* $x \in \mathbf{G}$ *and* $a \in \mathbf{N}$, *the conjugate* $x^{-1}ax$ *is in* $\mathbf{N}$.

Corollary 87. *Every subgroup of a commutative group is normal.*

Corollary 88. *Every subgroup of a group of index two is normal.*

Corollary 89. *The center of every group is a normal subgroup.*

Corollary 90. *The kernel of every homomorphism is a normal subgroup.*

When we form the quotient $\mathbf{G}/\mathbf{N}$ of $\mathbf{G}$ by a normal subgroup $\mathbf{N}$, we think of collapsing, and thereby eliminating, the elements of $\mathbf{N}$. It is sometimes possible to start with a group that is in some sense "bad", then gather the "bad" elements into a normal subgroup. By dividing out this normal subgroup, we get a quotient group that is "good" since it has no "bad" elements left.

For example, being commutative is surely a "good" property for a group. If $\mathbf{G}$ is a group and $a,b \in \mathbf{G}$, the **commutator** of a and b is defined as

$$[a,b] := a^{-1}b^{-1}ab.$$

Lemma 91. *For elements a,b,x in a group $\mathbf{G}$,*

(i) $[a,b] = e$ *if and only if a and b commute.*

(ii) $[a,b]^{-1} = [b,a]$ (*The inverse of a commutator is a commutator.*),

(iii) $x^{-1}[a,b]x = [x^{-1}ax, x^{-1}bx]$ (*The conjugate of a commutator is the commutator of the conjugates.*)

If you prefer commutative groups, then you have to think of commutators (other than e) as "bad" elements. Let $[\mathbf{G},\mathbf{G}]$ denote the set of all commutators and finite products of commutators of elements of $\mathbf{G}$.

Lemma 92. *If $\mathbf{G}$ is a group, then $[\mathbf{G},\mathbf{G}]$ is a normal subgroup of $\mathbf{G}$* (*called the* **commutator subgroup** *of $\mathbf{G}$*).

Because each commutator different from e witnesses the failure of two elements to commute, the size of the commutator subgroup is a measure of how non-commutative $\mathbf{G}$ is. Thus, by (i), $[\mathbf{G},\mathbf{G}] = \{e\}$ if and only if $\mathbf{G}$ is commutative, and a larger commutator subgroup indicates more non-commuting elements. When we form the quotient group $\mathbf{G}/[\mathbf{G},\mathbf{G}]$, we collapse all of the witnesses of failure of commutativity to the identity. The result is a group with no witnesses to failure of commutativity.

Theorem 93. *For every group $\mathbf{G}$, the quotient $\mathbf{G}/[\mathbf{G},\mathbf{G}]$ of $\mathbf{G}$ by its commutator subgroup is a commutative group.*

Another illustration of quotient groups comes from an extension of Problem 53 and the group $\mathbf{P}_{abc}$ with operation $\oplus$. Let $S = \{a,b,c,d,e,f,g,h\}$, $T = \{a,b,c\}$ and $U = \{d,e,f,g,h\}$. Problem 53 asks us to think of T as the "important" elements of S, and think of U as the "unimportant" elements of S. We then think of two members of $\mathbf{P}_S$ as being equivalent if they contain the same important elements.

Like $\mathbf{P}_{abc}$, the collection of subsets $\mathbf{P}_S$ of S, subsets $\mathbf{P}_T$ of T and subsets $\mathbf{P}_U$ of U each form a group under symmetric difference $\oplus$.

Problem 94. *Define $f : \mathbf{P}_S \to \mathbf{P}_T$ by $f(X) := X \cap T$. Thus $f(X)$ consists of the "important" elements of X.*

(i) *Show that f is a homomorphism from $\mathbf{P}_S$ onto $\mathbf{P}_T$.* (*This says that the sum of the important elements in X and the important elements in Y is exactly the important elements in $X \oplus Y$.*)

(ii) *Show that $\mathbf{P}_U$ is the kernel of f.*

(iii) *For $X,Y \in \mathbf{P}_S$, show that $f(X) = f(Y)$ if and only if X and Y are in the same coset of $\mathbf{P}_U$.*

(iv) *Show the* $\mathbf{P}_S/\mathbf{P}_U \cong \mathbf{P}_T$. (*Be sure that your isomorphism is well defined.*)

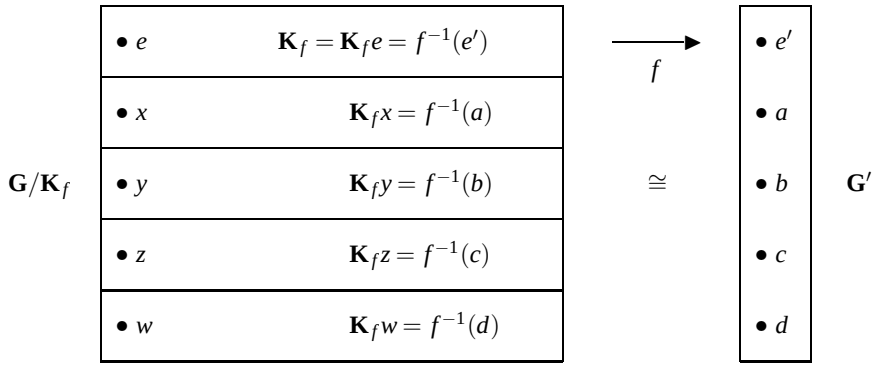
We will see that the conclusion of Problem 94 is quite general. A group $\mathbf{G}'$ is said to be a **homomorphic image** of a group $\mathbf{G}$ if there is a homomorphism $f : \mathbf{G} \to \mathbf{G}'$ from $\mathbf{G}$ onto $\mathbf{G}'$.

Theorem 95. *Every quotient of a group* $\mathbf{G}$ *is a homomorphic image of* $\mathbf{G}$. *More specifically, if* $\mathbf{N}$ *is a normal subgroup of* $\mathbf{G}$, *then the* **natural homomorphism** $h : \mathbf{G} \to \mathbf{G}/\mathbf{N}$, *given by* $h(x) := \mathbf{N}x$, *is a homomorphism from* $\mathbf{G}$ *onto* $\mathbf{G}/\mathbf{N}$.

Lemma 96. *Let* $f : \mathbf{G} \to \mathbf{G}'$ *be a homomorphism,* $x, y \in \mathbf{G}$. *Then* $f(x) = f(y)$ *if and only if* $\mathbf{K}_f x = \mathbf{K}_f y$.

We can now prove that every homomorphic image of a group is isomorphic to a quotient of that group.

The Isomorphism Theorem 97. *If* $f : \mathbf{G} \to \mathbf{G}'$ *is an onto homomorphism, then* $\mathbf{G}/\mathbf{K}_f \cong \mathbf{G}'$.



(For example, if $xy = w$, then $\mathbf{K}_f x \cdot \mathbf{K}_f y = \mathbf{K}_f w$ and $ab = d$.)

Wow – quotient groups – what a great idea! And in the end, they turn out to be exactly the same as homomorphic images. But just as you are thinking this, there is an unsettling disturbance.

"You – yes, you in the back there! Can you come up to the board and show us your table for $\mathbf{D}_4$?"

"Well, no Professor, I'm sorry – I can't. But I've just found a really neat way to patch together a whole bunch of new groups! Can I put that up instead?"